

PRESSENTATION

Data Visualization Trainee Early Remote Internship

Presentation

*Excelerate X Depaul University
International Admission Analytics*



Data Visualization Trainee

Early Remote Internship

(Excelerate X DePaul University)

ROLE: DATA ANALYST



4 weeks

Presented by

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Internship Objective

Purpose: The purpose is to develop practical skills and gain hands on experience in data visualization.

Goal: The goal is to communicate insights clearly and support better decision-making.

Approach: Data Extraction, Data Familiarization, Data Cleaning, Visualization, Insights and Recommendation

Data Extraction

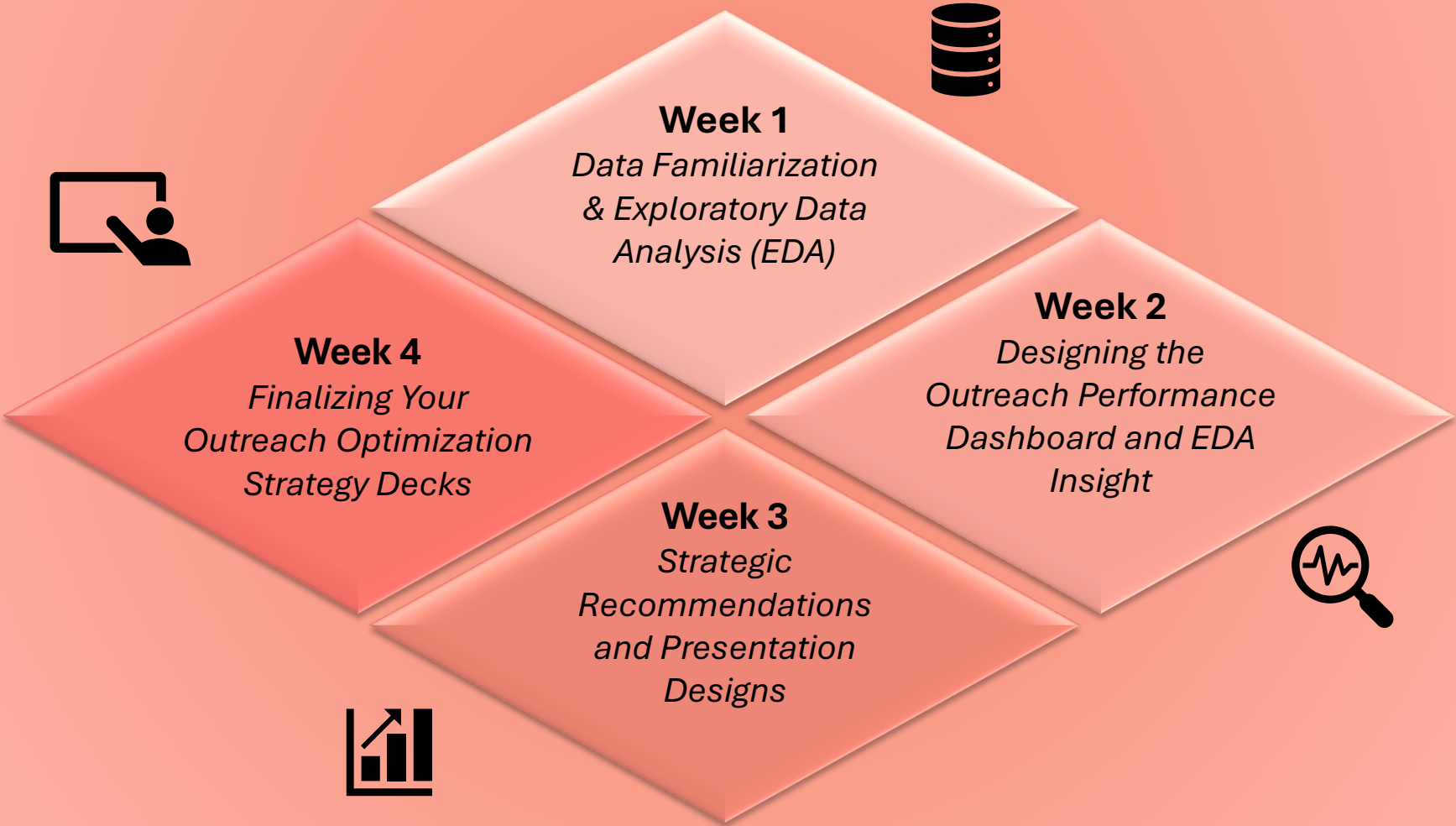
*Data
Familiarization*

*Data
Cleaning*

Visualization

*Insights and
Recommendat
ion*

Weekly Deliverables



Data Visualization Trainee

Early Remote Internship

 4 weeks

**Key
Takeaways at a
Glance**

Data Visualization Trainee

Early Remote Internship



4 weeks

Key Takeaways at a Glance

84% Non-Study Group

*Non-Study Group
channels drive the
majority of applications
and enrollment
conversions*

7,543 Applicant Records

*Comprehensive dataset
spanning 2023–2025
across 42 columns*

India: 54% of Applicants

*India is the dominant
source country, with
Hyderabad and
Telangana leading
regional output*

Computing & Business Lead

*Computer Science and
Business Analytics are the
most popular graduate
programs*

Data Familiarization and Data Dictionary

Applicants

- ❖ Reference ID
- ❖ Country
- ❖ College
- ❖ University
- ❖ Intake
- ❖ College 1st Choice
- ❖ Admit Date
- ❖ Application Source

Outreach

- ❖ Reference ID 2
- ❖ Caller Name

Campaign

- ❖ Campaign ID
- ❖ Slate Form Filled

Variable Name	Description	Data Type	Example
Reference ID	Unique Identifier of Applicants	Integer	45405320
Admit Date	The admission date	Date	20/02/2024
Country	Applicant's country of residence	Text	India
Region	Applicant's state or region	Text	Telangana
City	Applicant's city	Text	Hyderabad
1 st Choice College	Preferred college selected by applicant	Text	Jarvis College
Subject/Intake	Program Applied for	Text	Computer Science
Application source	Source of application	Text	Non-Study
Campaign ID	Recruitment Campaign Identifier	Integer	747758
Outreach Reference ID	Outreach activity Identifier	Integer	902518555
Caller Name	Counselor handling the applicant	Text	Jagdeep
Slate Form Filled	Slate application status	Text	Yes/No

Dataset Overview & Exploratory Data Analysis

The dataset contains 7,543 applicant records across 42 columns, covering applicant information, campaign data, outreach activities, and enrollment outcomes from 2023–2025. Week 1 EDA surfaced key patterns in geography, program demand, and recruitment channel performance.

Dataset Summary

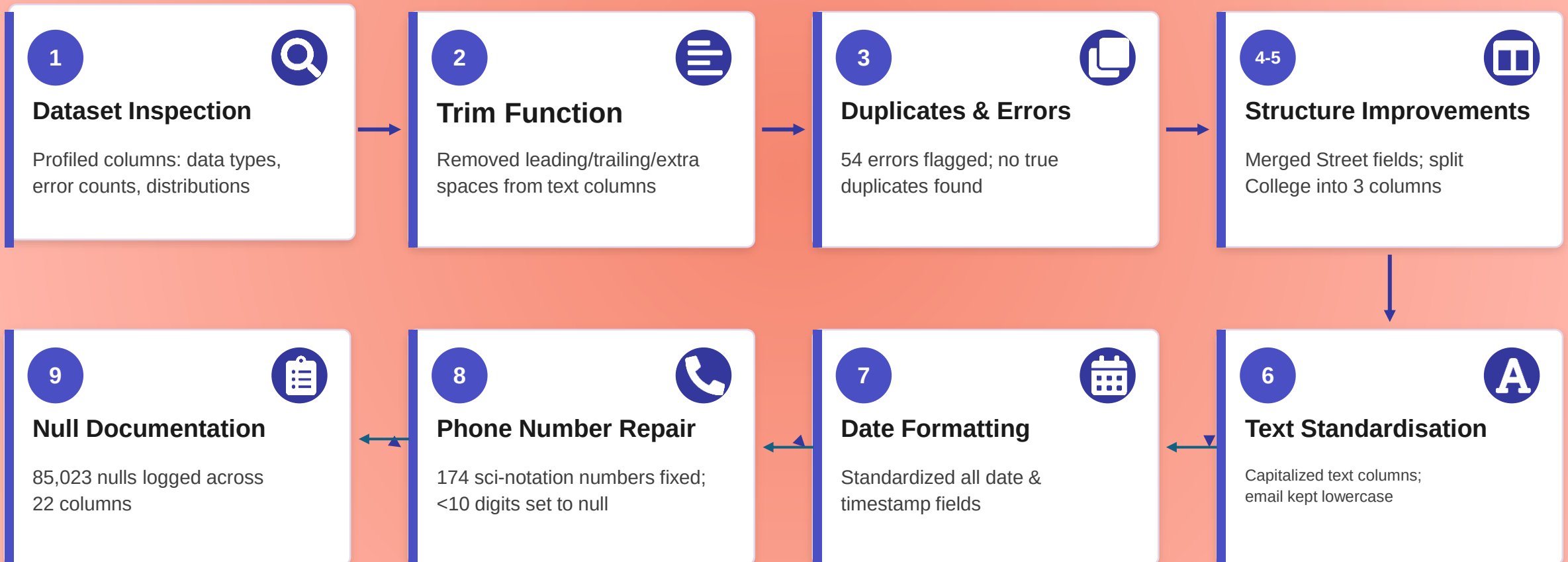
- **7,543** total records
- **42** data columns
- **2023–2025** coverage
- Graduate-level admissions
- Multiple source countries

Key EDA Findings

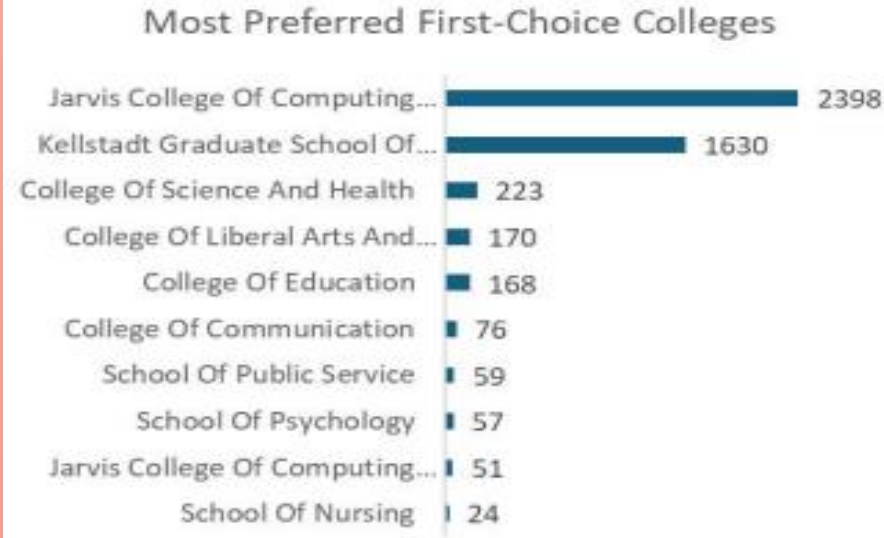
- India is the largest applicant source by a significant margin
- Jarvis College receives the highest volume of applications
- Non-Study Group channels dominate overall recruitment
- Missing values exist across several columns requiring attention
- Duplicate Reference IDs represent repeated applicant interactions, not true duplicates

Data Quality Assessment & Cleaning — Process Flow

Week 2 followed an 8-stage Power Query workflow to validate, standardize, and repair the raw applicant export before it moved into visualization.



Visualization Report

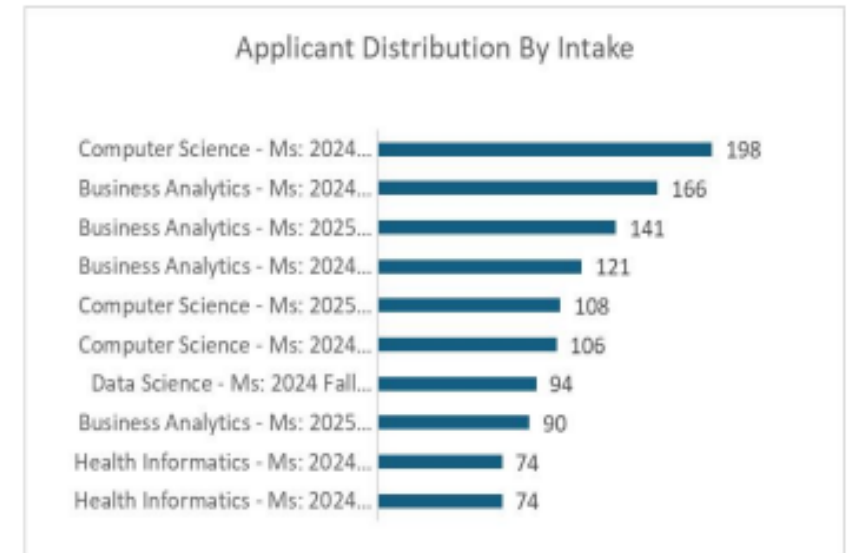
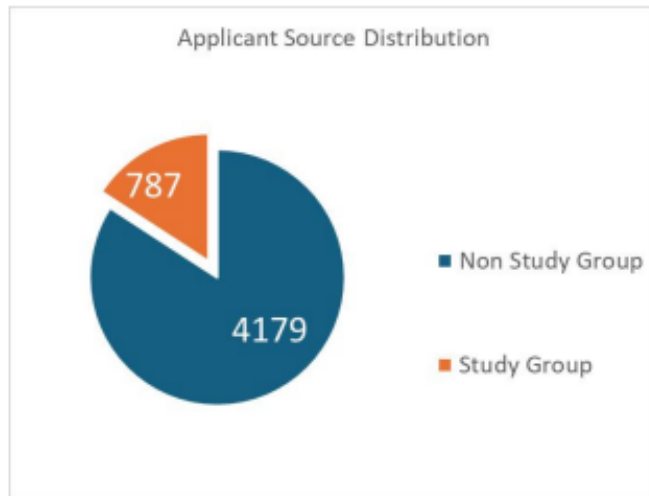


Most Preferred First-Choice Colleges

- Jarvis College of Computing and Digital Media ,it is the most preferred college (2,398 applicants).
- Kellstadt Graduate School of Business ranks second (1,630 applicants).

Top Applicant Countries

- India contributes the highest number of applicants (2,679), representing the largest international market.
- The United States (487), Ghana (409), and Nigeria (359) are the next major applicant sources.
- Emerging markets such as Pakistan, China, and Nepal also contribute to applicant diversity.



Application Source Distribution

- Non-Study Group: 4,179 applicants (84%)
- Study Group: 787 applicants (16%)
- The majority of applications are received through the Non-Study Group, making it the university's primary recruitment channel.

Top Applicant Cities

- Hyderabad: 570 applicants (Highest)
- Chicago: 184 applicants
- Accra: 140 applicants
- Ahmedabad: 139 applicants
- Mumbai: 128 applicants

Applicant Distribution by Intake

- Computer Science (MS – 2024): 198 applicants (Highest)
- Business Analytics (MS – 2024): 166 applicants
- Business Analytics (MS – 2025): 141 applicants
- Business Analytics (MS – 2024): 121 applicants
- Computer Science (MS – 2025): 108 applicants

Interactive Power BI Dashboard

The Week 3 deliverable is a fully interactive Power BI dashboard designed for admissions leadership and international recruitment managers. It combines KPI summaries, trend visualizations, and drill-down filters to enable fast, data-driven decision-making without requiring technical expertise.



KPI Cards & Gauge Charts

At-a-glance performance metrics including total applicants, unique applicants, campaign IDs, and outreach activity counts.



Area & Line Charts

Time-series views of application trends across admission years, highlighting growth patterns and seasonal fluctuations.



Donut & Treemap Charts

Visual breakdowns of applicant distribution by country, program, and recruitment channel for quick pattern recognition.



Funnel Analysis & Filters

Enrollment conversion funnel from application to enrollment, paired with interactive filters for segment-level exploration.

KPI Snapshot & Applicant Profile

7,543

Total Applicants

Across all programs and admission years

5,000

Unique Applicants

After deduplication of repeated interactions

4,349

Campaign IDs

Recruitment campaigns tracked with applicants reference Id

1,806

Unique Outreach IDs

Individual outreach activities recorded

Who Are Our Applicants?

- **India** contributes over 54% of total applicants — the single largest source country
- **Hyderabad** leads all cities in applicant volume, followed by other major Indian metros
- **Telangana** is the top-producing region, reflecting strong graduate program awareness
- **Computer Science** remains the most popular program, with **Business Analytics** ranking second
- **2024** was the peak admission year, signaling growing international interest

Recruitment Channel Performance

- **Non-Study Group** channels account for 84% of all applications
- Study Group channels contribute the remaining 16%
- Non-Study Group also leads in enrollment conversions
- Direct applications are the dominant application type
- Referral and agent channels show meaningful but smaller volumes

Program Demand, Enrollment & Outreach

Program-level analysis reveals clear demand concentration, while enrollment and outreach data highlight conversion dynamics and the campaign activity patterns that drive applicant engagement.

Program Demand

Computer Science leads in applications, followed by Business Analytics, Data Science, and Information Systems. This concentration signals strong market demand for technology and business programs, guiding targeted recruitment messaging and resource prioritization.

Enrollment Conversion

Approximately 66% of applicants convert to enrolled students. Non-Study Group channels demonstrate higher conversion rates, suggesting more effective applicant qualification or stronger program alignment within those recruitment pathways.

Outreach Activity

Outreach peaked in 2024, with email as the dominant contact method. Higher outreach volumes correlate with increased application rates, reinforcing the value of sustained, multi-touch engagement campaigns.



Key Insight: Aligning outreach volume with high-demand programs and top-performing channels can improve conversion efficiency and reduce cost-per-enrollment.

Strategic Findings & Recommendations

The analysis reveals four high-impact opportunities to strengthen DePaul's international graduate recruitment strategy. Each recommendation is grounded in data patterns identified across the EDA, data cleaning, and dashboard phases.

Focus on India

India accounts for over 54% of applicants. Concentrate recruitment resources, regional events, and localized marketing in Hyderabad and Telangana to maximize yield from the highest-potential market.

Strengthen Non-Study Group Channels

Non-Study Group drives 84% of applications and leads in enrollment conversions. Invest in deepening these partnerships and replicating their engagement models across other channels.

Promote High-Demand Programs

Computer Science and Business Analytics dominate applicant interest. Prioritize these programs in international marketing campaigns, scholarship offerings, and admissions communications.

Improve Data Collection

85,023 missing value instances highlight gaps in data capture. Implement standardized data entry protocols and validation rules to improve dataset completeness for future analysis cycles.

Conclusion & Next Steps

This project delivered a complete analytics pipeline — from data exploration and quality assurance to an interactive Power BI dashboard — providing DePaul's admissions leadership with a reliable foundation for data-driven recruitment decisions. The following next steps will extend the value of this work.

→ Expand Data Integration

Connect the dashboard to live admissions data sources to enable real-time monitoring of application volumes, campaign performance, and enrollment trends.

→ Enhance Regional Targeting

Use geographic and program-level insights to design hyper-targeted outreach campaigns for high-potential regions beyond India, diversifying the applicant pipeline.

→ Conduct Predictive Modeling

Build enrollment prediction models using historical applicant data to forecast yield rates, identify at-risk applicants, and optimize recruitment spend.

→ Establish Ongoing Reporting

Schedule quarterly dashboard reviews with admissions leadership to track KPIs, measure campaign ROI, and refine recruitment strategy based on emerging trends.

✅ **Impact:** This analytics solution positions DePaul to make faster, more confident decisions in international graduate recruitment — turning raw data into a competitive advantage.

THANK YOU

“Special Thanks to Kolawole Oparinde”